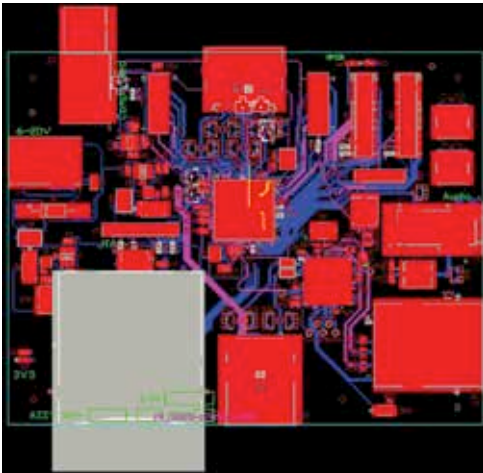


Prototype Pi to Present Day Pi

As we know the creators of Raspberry Pi were motivated by the lagging programming literacy in Britain. As alumni of the prestigious University of Cambridge, and as professionals in the computer industry, they recognised that this diminishing aptitude for core computing skills needed to be fixed. The purpose of the Pi was to make computing and programming an early development skill which could be used to educate children. The core features of the product they imagined was to be an open learning environment which could be bought for cheap by any school and parent. With that in mind they set about looking at the current landscape and decided to build a microcomputer which could form the basis of this learning experience.

The design and philosophy of the device was borrowed from 1981's BBC Micro computer made by Acorn Computers. But with the advantages



The Alpha design schematic is a work of art in itself.

of the miniaturisation of components accessible it was decided in advance that the Raspberry Pi would be about the size of a credit card. The lead designer of the project was Eben Upton, who was a chip architect at Broadcom, and was working to bring that form factor to life.

The early 2006 concept design was made using raw materials in a homemade

setting. It was initially hoped that users could assemble their own very cheap microcomputers at home. Using a veroboard base, Upton employed the Atmel ATmega644 microcontroller which clocked only 22.1 MHz with 512K of SRAM for data and frame buffer storage. The Atmel microcontroller only had 32 GPIOs, of which 19 were used to drive SRAM address bus needed to generate the video output. The video resolution was limited to 320x240 which the microcontroller juggled while performing other operations. Given